

Assignment 5

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1 BFS and DFS

1.a BFS

0, 3, 9, 10, 1, 4, 7, 2, 5, 6, 8

1.b DFS

0, 3, 1, 2, 6, 7, 9, 10, 8, 5, 4

2 Dijkstra's Algorithm

A -> A = 0

A -> B = 1

A -> C = 3

A -> D = 6

A -> F = 9

A -> C = 13

A -> C = 12 = winner (A-B-F-E)

3 Adjacency List and Matrix

3.a Adjacency List

A: B C D

B: A E

C: A E

D: A E

E: B C D F

F: D E

3.b Adjacency Matrix

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

4 Graph Algorithms

